

European Football Analytics

2025-2026 Season | Top 5 Leagues | Player Statistics

Data Science Internship Project

Decode Labs | All 5 Tasks Completed

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#	Task	Status
1	Data Collection & Dataset Understanding	Complete
2	Data Cleaning & Preprocessing	Complete
3	Exploratory Data Analysis (EDA)	Complete
4	Data Visualization (10 Charts)	Complete
5	Predictive Model + Performance Score	Complete

GitHub Repository: github.com/mohamedmamdouh342-hash/decode-labs-data-science-internship

1. Dataset Overview

The dataset used for this project is the European Football Players Statistics 2025-2026, sourced from FBref (Football Reference). It covers player performance data from the top five European football leagues across the 2025-2026 season, providing a comprehensive view of attacking, defensive, and goalkeeping contributions at the player level.

Total Players	2,779	Total Columns	53
Season	2025-2026	Leagues	5
After Cleaning	2,439	Missing Values After Cleaning	0

Dataset Column Categories

Category	Columns	Examples
Player Identity	8	Player, Nation, Pos, Squad, Comp, Age, Born
Playing Time	4	MP, Starts, Min, 90s
Attacking	7	Gls, Ast, G+A, G-PK, PK, PKatt, G+A-PK
Shooting	8	Sh, SoT, SoT%, Sh/90, SoT/90, G/Sh, G/SoT
Discipline	8	CrdY, CrdR, 2CrdY, Fls, Fld, OG
Defending	3	TkIW (Tackles Won), Int (Interceptions), Crs (Crosses)
Goalkeeping	13	GA, GA90, SoTA, Saves, Save%, W, D, L, CS, CS%, PKA, PKsv, PKm

Players per League

League	Players	Total Goals
Premier League	479	924
La Liga	515	899
Bundesliga	439	908
Serie A	531	824
Ligue 1	475	788

2. Data Cleaning & Preprocessing

The raw dataset required systematic cleaning before analysis. Five cleaning steps were applied to ensure data quality, consistency, and analysis readiness.

Step	Action	Result
1	Remove duplicate player entries (mid-season transfers)	2,779 → 2,459 rows
2	Filter players with < 90 minutes played	2,459 → 2,439 rows
3	Standardize positions (MF,FW → MF primary)	4 clean roles: GK/DF/MF/FW
4	Clean league names (remove country prefixes)	Premier League, La Liga, etc.
5	Fill missing values (numeric → 0, text → Unknown)	0 remaining NaN cells

Position Distribution After Cleaning

Position	Count	Percentage
Midfielder (MF)	1,102	45.2%
Defender (DF)	749	30.7%
Forward (FW)	413	16.9%
Goalkeeper (GK)	175	7.2%

3. Exploratory Data Analysis

3.1 Descriptive Statistics

Stat	Age	Minutes	Goals	Assists	Shots	TkIW	Int
Mean	26.15	1,305	1.78	1.24	16.55	12.79	10.91
Median	26.00	1,206	1.00	1.00	11.00	10.00	7.00
Max	42.00	3,150	33.00	19.00	132.00	67.00	73.00
Min	16.00	90	0.00	0.00	0.00	0.00	0.00

Key insight: The median goals per player is 1 while the mean is 1.78 — indicating a right-skewed distribution where a small number of elite scorers pull the average up. Over 40% of outfield players score 0 or 1 goal in an entire season.

3.2 Goals by League

League	Total Goals	Avg Goals per Player
Bundesliga	908	2.07
Premier League	924	1.93
La Liga	899	1.75
Ligue 1	788	1.66
Serie A	824	1.55

The Bundesliga leads in average goals per player (2.07), suggesting a more open, attacking style of play compared to Serie A which has the lowest average (1.55).

3.3 Average Stats by Position

Position	Goals	Assists	Shots	SoT	TkIW	Int	Crosses
Forward (FW)	4.22	1.38	29.42	11.73	5.82	3.12	13.50
Midfielder (MF)	1.89	1.68	19.38	6.37	14.71	11.09	33.74
Defender (DF)	0.69	0.79	9.15	2.60	16.73	17.43	18.57
Goalkeeper (GK)	0.00	0.07	0.05	0.01	0.29	0.22	0.01

Position profiles are clearly distinct. Forwards lead in goals and shots. Midfielders lead in crosses and assist in both attacking and defensive phases. Defenders dominate in interceptions and tackles. Goalkeepers are statistically unique — zeros across all outfield metrics — which makes them the easiest position for the classifier to identify.

3.4 Age and Performance

Age Group	Avg Goals (Outfield)
Under 20	1.37
20 - 22	1.27
23 - 25	2.10
26 - 28	2.12
29 - 31	2.26
32 - 34	1.86
35+	1.94

Goal contributions peak in the 26-31 age range (avg 2.12-2.26 per player), confirming the widely held football belief that players reach their prime between ages 26 and 31. Young players (U22) show lower averages as they are still developing.

3.5 Top Performers — Attacking

Top 5 Goal Scorers

Player	Club	League	Goals	Shots	SoT
Harry Kane	Bayern Munich	Bundesliga	33	111	63
Erling Haaland	Manchester City	Premier League	25	119	54
Kylian Mbappe	Real Madrid	La Liga	24	132	61
Igor Thiago	Brentford	Premier League	22	79	41
Vedat Muriqi	Mallorca	La Liga	21	97	44

Top 5 Assist Providers

Player	Club	League	Assists	Goals
Bruno Fernandes	Manchester Utd	Premier League	19	8
Michael Olise	Bayern Munich	Bundesliga	19	14
Federico Dimarco	Inter	Serie A	16	6
Julian Ryerson	Dortmund	Bundesliga	14	0
Luis Diaz	Bayern Munich	Bundesliga	13	15

3.6 Most Clinical Finishers (G/Sh, minimum 10 shots)

Player	Club	League	Goals	Shots	G/Sh
Raphael Guerreiro	Bayern Munich	Bundesliga	5	11	0.45
Orri Steinns Oskarsson	Real Sociedad	La Liga	6	15	0.40
Hugo Alvarez	Celta Vigo	La Liga	4	11	0.36
Taiwo Awoniyi	Nott. Forest	Premier League	4	12	0.33
Roberto Navarro	Athletic Club	La Liga	6	18	0.33

G/Sh (Goals per Shot) reveals finishing quality independently of playing time or chance volume. Guerreiro converted 45% of his shots — an elite-level finishing rate.

3.7 Top Defensive Players (TklW + Int)

Player	Club	League	Position	TklW	Int	Total
James Garner	Everton	Premier League	MF	67	53	120
Malang Sarr	Lens	Ligue 1	DF	42	73	115
Victor Nelsson	Hellas Verona	Serie A	DF	37	70	107
Jon Aramburu	Real Sociedad	La Liga	DF	60	43	103

Nicolas Seiwald	RB Leipzig	Bundesliga	MF	43	60	103
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3.8 Top Goalkeepers by Save%

Player	Club	League	GA	Saves	Save%	Clean Sheets
Janis Blaswich	Leverkusen	Bundesliga	9	41	82.0%	3
Joan Garcia	Barcelona	La Liga	20	72	80.0%	14
Mike Maignan	Milan	Serie A	29	94	79.3%	14
Ivan Provedel	Lazio	Serie A	27	87	78.6%	12
Mile Svilar	Roma	Serie A	29	101	78.3%	16

4. Data Visualization

Ten professional charts were produced using matplotlib and seaborn, each designed to answer a specific football question clearly. All charts are saved in the visuals/ folder of the GitHub repository.

#	Chart Title	Football Question Answered
1	Top 15 Goal Scorers	Who scored the most goals this season?
2	Top 15 Assist Providers	Who created the most for teammates?
3	Goals Distribution by League	How are goals distributed across leagues?
4	Player Position Donut Chart	What is the composition of players by role?
5	Age Distribution Histogram	Where is the prime age zone?
6	Goals vs Assists Scatter	Who scores AND creates? Who specializes?
7	Correlation Heatmap	Which stats are related to each other?
8	Normalized Stats by Position	How does each position differ statistically?
9	Top 12 Defensive Players	Who dominated defensively this season?
10	Top 10 Goalkeepers by Save%	Who saved the most percentage of shots?

All charts use league-specific brand colors (Premier League purple, La Liga red, Bundesliga red, Serie A blue, Ligue 1 dark blue) and position-specific green shades for immediate visual identification. Charts are saved at 150 DPI for high-resolution output.

5. Predictive Model

5.1 Model Overview

A Random Forest Classifier was trained to predict a player's position (FW, MF, DF, GK) using only their season statistics — no position label used as input. This mirrors how real football scouting databases categorize players from raw data.

Parameter	Value
Algorithm	Random Forest Classifier
Number of Trees	200
Max Tree Depth	12
Min Samples Leaf	3
Class Weighting	Balanced (handles GK imbalance)
Train/Test Split	80% / 20% (1,951 train / 488 test)
Feature Scaling	StandardScaler
Overall Accuracy	75.41%

5.2 Features Used (14 total)

Category	Features
Attacking	GlS, Ast, G+A, Sh, SoT
Defending	TkIW, Int, Crs
Goalkeeping	GA, Saves, Save%, CS
Context	Min (minutes played), Age

5.3 Classification Results

Position	Precision	Recall	F1-Score	Support
DF (Defender)	71%	77%	74%	150
FW (Forward)	75%	72%	74%	83
GK (Goalkeeper)	100%	100%	100%	35
MF (Midfielder)	75%	72%	73%	220
Overall Accuracy	75.41%			488

5.4 Confusion Matrix Analysis

Actual \ Predicted	DF	FW	GK	MF
DF	115	2	0	33

FW	2	60	0	21
GK	0	0	35	0
MF	44	18	0	158

Goalkeepers are classified with 100% accuracy — their stats (Saves, GA, Save%, CS) are uniquely non-zero only for the 175 goalkeepers in the dataset. The primary confusion is between Defenders and Midfielders (33 DF misclassified as MF, 44 MF as DF) — which is a real football observation: defensive midfielders share similar tackle and interception profiles with central defenders.

5.5 Feature Importance

Rank	Feature	Importance	Football Meaning
1	Int (Interceptions)	14.09%	Separates DF from FW/MF
2	Sh (Shots)	10.87%	Separates FW from DF/GK
3	Saves	10.37%	Perfect GK identifier
4	Save%	10.37%	Perfect GK identifier
5	Crs (Crosses)	9.81%	Differentiates MF from FW
6	TkIW (Tackles Won)	8.67%	Separates DF/MF from FW
7	SoT (Shots on Target)	7.59%	Separates FW from others
8	Min (Minutes)	6.09%	Context — affects all stats

5.6 Performance Score System

A custom weighted scoring formula was built to fairly rank players within each position. Rather than a generic total-stats score, each position is scored based on what their role specifically demands in football.

Position	Scoring Formula	Primary Weight Reason
Forward (FW)	Goals x4 + Assists x2 + SoT x0.5 + Sh x0.2	Goals are the striker's primary job
Midfielder (MF)	Goals x2.5 + Assists x2.5 + TklW x0.8 + Int x0.8	Balance across both phases
Defender (DF)	TklW x2.5 + Int x2.5 + Goals x1.5 + Assists x1.0	Defensive actions are primary
Goalkeeper (GK)	Saves x1.5 + Clean Sheets x2.0 + Save% x0.3	Keeping goals out is the job

Top 10 Forwards by Performance Score

Player	Club	League	Age	Score
Harry Kane	Bayern Munich	Bundesliga	32	195.7
Erling Haaland	Manchester City	Premier League	25	164.8
Kylian Mbappe	Real Madrid	La Liga	27	160.9
Deniz Undav	Stuttgart	Bundesliga	29	128.7
Vedat Muriqi	Mallorca	La Liga	32	127.4
Igor Thiago	Brentford	Premier League	24	126.3
Esteban Lepaul	Rennes	Ligue 1	26	113.7
Vinicius Junior	Real Madrid	La Liga	25	111.0
Lautaro Martinez	Inter	Serie A	28	109.4
Ante Budimir	Osasuna	La Liga	34	104.1

Top 10 Midfielders by Performance Score

Player	Club	League	Age	Score
James Garner	Everton	Premier League	25	118.5
Nicolas Paz	Como	Serie A	21	109.8
Bruno Fernandes	Manchester Utd	Premier League	31	107.5
Federico Dimarco	Inter	Serie A	28	103.0
Michael Olise	Bayern Munich	Bundesliga	24	102.5
Christoph Baumgartner	RB Leipzig	Bundesliga	26	96.4
Luis Diaz	Bayern Munich	Bundesliga	29	94.0
Moises Caicedo	Chelsea	Premier League	24	90.8
Mateus Fernandes	West Ham United	Premier League	21	90.2

Lamine Yamal	Barcelona	La Liga	18	89.9
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Top 10 Defenders by Performance Score

Player	Club	League	Age	Score
Malang Sarr	Lens	Ligue 1	27	288.5
Victor Nelsson	Hellas Verona	Serie A	27	267.5
Jon Aramburu	Real Sociedad	La Liga	23	259.0
Neco Williams	Nott. Forest	Premier League	25	248.5
Oumar Solet	Udinese	Serie A	26	238.0
Antonio Caracciolo	Pisa	Serie A	35	235.5
Carlos Romero	Espanyol	La Liga	24	234.5
Iglesias	Getafe	La Liga	27	233.5
Adrien Truffert	Bournemouth	Premier League	24	227.0
David Affengruber	Elche	La Liga	25	220.0

Top 10 Goalkeepers by Performance Score

Player	Club	League	Age	Score
Aaron Escandell	Oviedo	La Liga	30	243.96
Moritz Nicolas	Gladbach	Bundesliga	28	235.11
Emil Audero	Cremonese	Serie A	29	218.43
Martin Dubravka	Burnley	Premier League	37	218.33
Marko Dmitrovic	Espanyol	La Liga	34	218.13
Herve Koffi	Angers	Ligue 1	29	212.66
Arijanet Muric	Sassuolo	Serie A	27	211.65
Mile Svilar	Roma	Serie A	26	206.99
Elia Caprile	Cagliari	Serie A	24	206.74
Mathew Ryan	Levante	La Liga	34	203.26

6. Key Findings & Conclusion

Key Findings

Goal-scoring is extremely rare

Median goals = 1 across all outfield players. Over 40% of outfield players score 0 or 1 goal per season. Only a tiny elite tier of ~30 players reaches double figures.

Harry Kane dominates Europe

Kane leads all scorers with 33 goals in the Bundesliga — 8 more than second-placed Haaland (25). His shots-on-target rate (57%) is the highest among top scorers.

Bundesliga is the highest-scoring league

The Bundesliga averages 2.07 goals per player — the highest of all five leagues. Serie A has the lowest average at 1.55 goals per player.

Peak performance age is 26-31

Average goal contributions peak in the 26-31 age bracket (2.12-2.26 goals per player). Players under 22 average 1.27-1.37, confirming they are still in development.

Goalkeepers are perfectly classifiable

The model achieves 100% accuracy on goalkeepers. Saves, GA, Save%, and CS are uniquely non-zero only for the 175 goalkeepers — separating them perfectly.

DF vs MF is the hardest classification

33 Defenders were misclassified as Midfielders and 44 Midfielders as Defenders. This reflects a real football reality — defensive midfielders share tackle and interception profiles with central defenders.

Bruno Fernandes and Michael Olise lead assists

Both recorded 19 assists — the highest in Europe this season. Fernandes contributed mainly through creative passing while Olise combined 14 goals with his 19 assists.

Interceptions are the strongest position identifier

Feature importance analysis shows Int (interceptions) is the single most important feature (14.09%) for predicting position — more important than goals or saves.

Project Summary

Task	Deliverable	Status
1 — Data Collection	Dataset loaded, 53 columns documented	Complete
2 — Data Cleaning	Clean dataset, 2,439 players, 0 missing values	Complete
3 — EDA	Goals, shooting, age, defending, GK analysis	Complete
4 — Visualization	10 professional charts saved to visuals/	Complete
5 — Predictive Model	Role classifier (75.41%) + Performance Score	Complete

Technologies Used

Library	Version	Purpose
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pandas	2.0+	Data loading, cleaning, and analysis
numpy	1.24+	Numerical operations and array handling
matplotlib	3.7+	Chart drawing and customization
seaborn	0.12+	Statistical visualizations (boxplot, heatmap)
scikit-learn	1.3+	Machine learning model and evaluation
jupyter	1.0+	Interactive notebook environment